

Course Outline

CPSC 4830 001: Data Mining for Data Analytics

Fall 2025

Course Format: Lecture/Lab 2.0h (Wed (B010) /Mon (T215)) – 16.30 to 18.20

Credits: 3

Course Description, Prerequisites, and Corequisites:

Once data has been gathered, it must be cleaned, processed, and analyzed to find the most appropriate model to describe the underlying data. Using text classification and anomaly detection, students identify and implement the phases of a data mining process to mine and extract frequent pattern and outliers within data.

Prerequisite(s): A minimum "C" grade in CPSC 4810, 4820, and DANA 4830.

Learning Outcomes:

- Interpret the contribution of Data Mining to the decision-support level of organizations. Identify and interpret informative patterns and trends in data.
- Understand complexity of Data Mining algorithms and their limitations.
- Categorize and carefully differentiate between situations for applying different Data Mining techniques: frequent pattern mining, regression, association, correlation, classification, prediction, filtering, clustering, anomaly, and outlier analysis.
- Be capable of confidently applying common Data Mining algorithms in practice and implementing their own for a real-world dataset.
- Evaluate the performance of different Data Mining algorithms.
- Propose Data Mining solutions for different scenario and applications.

Instructor(s): Chinmaya Mahapatra

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Office Hours: Mon/Wed from 2.30 PM to 03.30 PM *

*By Appointment Only (Online or Zoom). Email your queries with a brief problem description and put "CPSC 4830" in subject line. According to the queries, zoom meeting will be scheduled, or a call will be scheduled.

PS: Instructor will try to return student emails ASAP. But student should expect to wait for up to 24 hours for a reply. During weekends replies may be delayed by a day or two. Except an emergency, replies during weekend should not be expected.

Textbook and Course Materials (Suggested):

- 1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3rd Edition (Required)**
By: Aurélien Géron
Publisher: O'Reilly Media, Inc.
- 2. Transformers for Natural Language Processing and Computer Vision - 3rd Edition**
By : Denis Rothman
Publisher: O'Reilly Media, Inc.

Online References Reading:

- 1. Aurélien Géron – GitHub Link with python codes.**
<https://github.com/ageron?tab=repositories>
- 1. An Introduction to Statistical Learning**
<http://faculty.marshall.usc.edu/gareth-james/ISL/>
- 2. Coursera – Data Mining**
<https://www.coursera.org/specializations/data-mining>
- 3. Google Machine Learning Course**
<https://developers.google.com/machine-learning/crash-course>

Online Popular Open Data Repositories:

- 1. UC Irvine Machine Learning Repository**
<http://archive.ics.uci.edu/ml/>
- 2. Kaggle datasets**
<https://www.kaggle.com/datasets>
- 3. Amazon's AWS datasets**
<https://aws.amazon.com/fr/datasets/>

Required Hardware:

Students need to bring his/her own laptop (Windows, Mac, or Linux) to almost every class with at least 8GB of RAM (16GB recommended), Decent CPU (i5 and above, Latest MAC) and at least 100GB of SSD storage space.

Required Software: (Installation tutorial will be provided in class)

Python 3 (Jupyter Notebooks/PyCharm/VS Code) and Packages such as TensorFlow, Keras, Pandas, Scikit-Learn, Matplotlib, Seaborn, Numpy, Scipy.

Assessments and Weighting:

Assignments (2)	20%	*In order to get a C or higher grade in a computer science course, a student MUST achieve at least a 50% average in the exam components of the course.
Project	20%	
Midterm	30%	
Final Exam	30%	
Total	100%	

Grading:

GP	Grade	Average	Description
4.33	A+	90 – 100	Distinguished Achievement
4.00	A	85 – 89	
3.67	A-	80 – 84	
3.33	B+	76 – 79	Above Average Achievement
3.00	B	72 – 75	
2.67	B-	68 – 71	
2.33	C+	64 – 67	Satisfactory Achievement
2.00	C	60 – 63	
1.67	C-	55 – 59	
1.00	D	50 – 54	Marginal Performance
0.00	F	< 50%	Unsatisfactory
0.00	N		Did not complete
-	W		Withdrawal

Detailed Course Schedule:

Week	Lecture	TODO
1	Introduction What is Data Mining? (Definition and examples) Data Mining vs Machine Learning vs Artificial Intelligence	
2 & 3	Introduction to Regression Analysis Linear and Logistic Regression. Model Evaluation. Bias-Variance Trade-off, Model Fitting, Model Selection and Cross-Validation	Implement Linear and Logistic regression in Python
3,4 & 5	Classification Decision Tree, Random Forest, XG Boost, Imb. Classification Dimensionality Reduction Feature Selection and Extraction. PCA	Implementing Classification Algorithms in Python Performing PCA in Python Assignment 1
5, 6, 7, 8, 9	Introduction to Deep Learning TensorFlow and Keras Introduction. Convolutional Neural Networks (CNNs),	Implementing Deep Learning using its Packages in Python (TensorFlow and Keras) Assignment 2

10, 11 & 12	Natural Language Processing Explore Generative AI and Large Language Models with Hugging Face, ChatGPT	Using Natural Language Processing in Classification Midterm (Nov 11)
13	Cluster Analysis K-Means and Hierarchical Clustering	Cluster Analysis in Python
13	Project Presentations	
	Final Exams: Dec 03rd – 15th	

As a student at Langara, you are responsible for familiarizing yourself and complying with the following policies:

College Policies:

[E1003 - Student Code of Conduct](#)

[F1004 - Code of Academic Conduct](#)

[E2008 - Academic Standing - Academic Probation and Academic Suspension](#)

[E2006 - Appeal of Final Grade](#)

[F1002 - Concerns about Instruction](#)

[E2011 - Withdrawal from Courses and Deferred Standing](#)

Departmental/Course Policies:

Attendance: To do well in this course, you need to attend promptly and regularly. If you miss a lab, you are responsible for completing any missed activities. I also advise you to ask someone who was in class to teach you the work we did—this benefit both of you. Normally a score of zero will be given for missed exams, labs, tests, etc. Makeup exams will not be permitted, except in circumstances of serious, well-documented illness or injury, or the death of a close family member.

D2L Brightspace: We will use D2L as a repository for lecture information, course handouts, lab and assignment submissions, and for links to supplementary materials. You will find help at <http://iweb.langara.bc.ca/online-learner/d2l/get-connected-to-d2l/>

Due Dates: Students should submit their assignments/lab **on or before the announced due time and date**. An assignment will be accepted as **late for 24 hours** after the due date and due time. The penalty for submitting late is 20% i.e. the mark becomes 80% of what the work would have received. After that date the grade will be 0 for that assignment/lab. You will be given a variety of assignments. Lab assignments are designed to be completed in one or two scheduled lab sessions. Any additional time required will have to be done during open lab time or at home.

Only submit through D2L BrightSpace. Email submissions will not be accepted in any circumstances.

Plagiarism is not tolerated under any circumstances.

- **Plagiarism is falsely presenting another person's work or ideas as your own, using another person's work or ideas without acknowledging the source of the material, or allowing another person to do the same with your work.**
- You are encouraged to discuss concepts with classmates, but the work that you submit must be your own. I recommend that if you are discussing an assignment with someone, that you not write/type anything about it for an hour after talking, AND you acknowledge him or her as a source.

Any plagiarized work automatically receives a **grade of zero** and is referred to the [Office of Student Conduct and Judicial Affairs](#), which could result in a failing grade for the course, or the student being suspended or expelled from the College. This excerpt on Plagiarism was borrowed from Jeremy Hilliker's Course Outline.